

AiSee++: An Open-Source Rapid Prototyping Platform for Screen-free Heads-Up Computing

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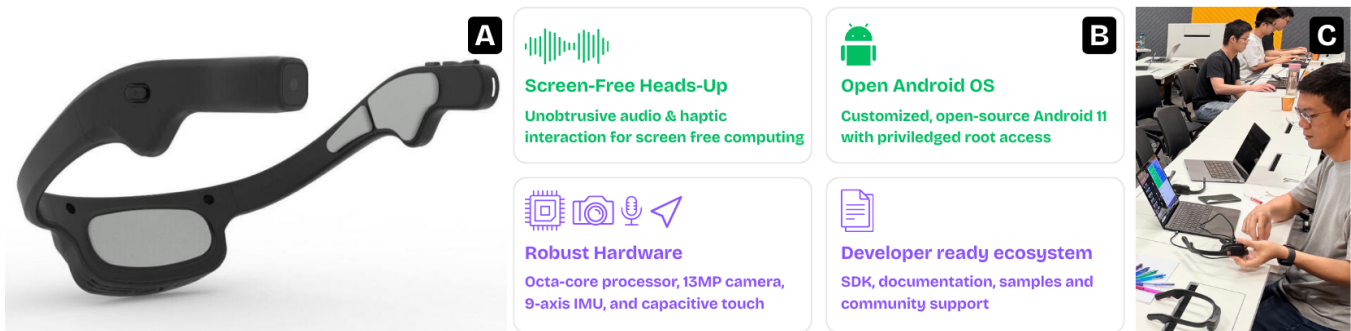


Figure 1: (A) The screen-free, head-worn device. (B) System highlights. (C) Workshop participants building functional prototypes.

Abstract

Heads-up computing aims to support everyday tasks through wearable technologies that enable natural, always-available interaction without disrupting attention. Yet, existing platforms often constrain how developers explore such interactions due to closed hardware, limited sensing, and restrictive software environments. We present a screen-free, head-worn prototyping platform with an open Android-based operating system and developer tools. In a 10-participant study, developers built functional prototypes within hours, demonstrating that the platform lowers technical barriers and enables rapid development of complex wearable applications. Also, our evaluation shows positive usability and developer experience, and identifies key opportunities for improvement. This work contributes an open wearable prototyping platform and empirical insights that inform future directions in screen-free heads-up computing.

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CCS Concepts

• **Human-centered computing** → **Interaction design**; **Human computer interaction (HCI)**; **Accessibility**;

Keywords

Heads-up Computing; Wearable computing; Prototyping tools;

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1 Introduction & Related Work

Computing tools now play a critical role in supporting everyday tasks, with many devices and solutions seamlessly integrated into modern life. In their article on “Heads-Up Computing”, Zhao et al. [24] highlight the importance of designing technology that aligns with natural human behaviour, positioning wearable computing devices as a key ingredient in realising this vision. One of the main characteristics that define heads-up computing is body compatible hardware modules that distribute device I/O modules to match that

of the human body [24]. Head-worn devices leverage the head as a rich site for sensing and interaction, and have become an important modality for heads-up computing. Our aim is to empower heads-up computing enthusiasts by introducing an open source platform for rapid prototyping of heads-up computing applications.

Display-equipped head-worn platforms such as Google Glass and Microsoft HoloLens have supported the development of heads-up computing applications [8, 18, 19]. However, near-eye visual overlays can divert attention from the surrounding environment and increase cognitive load [13, 22]. Near-eye displays may also induce visual discomfort and fatigue [9], motivating a shift toward screen-free head-worn devices for eyes-free interaction.

A natural screen-free alternative is ‘earables’, which expand earphones with sensing and interaction capabilities, are devices that are used continuously in everyday settings [6, 10]. These devices leverage audio and voice I/O, reducing visual distraction and aligning with the principles of heads-up computing. However, most of these devices are built primarily for only audio input with limited processing power and closed software that limit development of multimodal context aware applications [11, 23]. Open-source research platforms such as OpenEarable improve developer access, but remain focused on ear-centric sensing [20].

Many commercial head-worn devices remain closed platforms. For example, Ray-Ban Meta glasses primarily support third-party development through the Meta Wearables Device Access Toolkit, where developers are not provided access to run custom apps directly on the glasses [16]. In contrast, Mentra positions MentraOS as an open-source smart glasses platform, but its application model executes third-party apps in the cloud, which limits on-device programmability and direct access to device capabilities [14]. Project Aria provides a rich sensor suite, but it is a research-only kit and data access is mediated through the ARK software stack (companion app and client SDK), limiting broad developer access [7].

To address these limitations, we describe our design and development of a smart wearable with a headphone-based form factor and an open Android-based software stack.

Our device is designed around three features: high fidelity hardware for realistic and reproducible testing, an open and extensible OS that supports familiar Android development workflows, and strong developer resources including documentation and sample apps that accelerate iteration.

These features were derived from prior work analysis (see Section 2.1) as well as interviews with potential users. We conducted a workshop with 10 heads-up computing enthusiasts to evaluate our prototyping platform. In a 6-hour design sprint, participants with no prior experience using the platform were able to build functional prototypes. Our contributions include:

- Novel artifact: rapid prototyping platform that include a head-worn device featuring a novel form factor which runs an open source Android operating system. We also provide a development platform, including an SDK, example applications, and documentation to support rapid prototyping.
- Empirical findings: We share insights on platform usability, sample applications created by the developers and the developer experience.

2 System Design

2.1 Design Requirements

Previous work emphasises that heads-up computing devices must prioritise body-compatible form factors that align with natural human behavior to support seamless integration into daily life [24]. Furthermore, effective screen-free interaction requires multimodal, eyes-free interfaces leveraging spatial audio and subtle gestures that minimise cognitive load while maintaining social acceptability in public settings [3, 21]. Building on this literature, we derived an initial set of system requirements and conducted unstructured interviews to validate and refine them. We interviewed 10 participants (7 male, 3 female) with prior experience developing heads-up computing applications for other platforms. Participants came from various backgrounds, including developing these applications for both academic research as well as commercial products. The interviews were recorded, and our thematic analysis [2] revealed the following 3 themes.

[DR1] Robust hardware: Participants often encounter difficulties when prototyping with heads-up computing devices as they require knowledge of device configuration, hardware customisation and communication protocols. With growing interest in heads-up computing, wearable toolkits must be easy to use and support reliable deployment in time-constrained environments, while maintaining the flexibility for building sophisticated projects.

[DR2] Open and Extensible Software Stack: Most participants highlighted the need for fully open-source software with low-level access. They noted that while devices like the Meta Ray-Ban offer an SDK [17], the closed core OS prevents them from running custom applications directly on the device, limiting their ability to modify system behavior.

[DR3] Developer Resources: Proper documentation is a must, along with sufficient resources such as tutorials and examples that allow fast and effective testing. Necessary abstractions must also be in place to allow effective usage.

2.2 Hardware

Form Factor Design: Heads-up computing emphasises seamless and unobtrusive interaction that enables users to maintain attention on their surrounding environment [24]. Prior work on earables highlights ear-worn devices as a viable and increasingly adopted platform for always-available, eyes-free interaction [1, 10]. Motivated by these insights, we adopt a novel form factor for our device. As seen in Figure 2A, our device adopts a standalone behind-the-ear earable form factor with two lightweight arms, each ending in a forward-facing camera and privacy LED. The main modules are positioned toward the back of the head to keep the design screenless, providing an unobstructive experience while remaining comfortable and well balanced when worn (Figure 1A).

Processor & I/O: The device is powered by a MediaTek MT6762 octa-core processor operating at 2.0 GHz, paired with 4 GB of RAM and 64 GB of internal storage. Wireless connectivity includes dual-band WiFi (2.4 and 5.0 GHz), Bluetooth 5.0, and GPS with BeiDou support. Motion and orientation sensing are provided by a 9-axis

inertial measurement unit comprising a 3-axis accelerometer, gyroscope, and magnetometer. The hardware integrates a 13 MP IMX258 auto-focus camera with a 74 degree field of view, a single-channel microphone, and dual-channel open-ear speakers. User input and feedback are supported through four programmable tactile switches, a capacitive touch interface supporting swipe, single tap, double tap, and long press gestures, a vibration motor for haptic feedback, and a privacy LED. Power is supplied by an integrated 1100 mAh lithium-polymer battery, with charging and data transfer handled via a USB Type-C port (Fig. 2A).

2.3 Software

Operating system: Since open and extensible software was a major requirement [DR2], we developed a customised build of Android 11 as the operating system for our platform. Android offers a strong baseline for extensibility because Android Open Source Project (AOSP)¹ is publicly available for modification and redistribution on custom hardware deployments. It also benefits from broad developer familiarity and tooling due to its global adoption². Finally, Android provides mature libraries for heads-up sensing pipelines, such as CameraX for camera integration.³

Low-level access: To support rapid hardware prototyping, we customised our Android build to (1) expose GPIO and UART interfaces via privileged system components for direct serial communication, and (2) implement boot-time hooks that automatically start trusted accessibility services, which capture device events, such as key presses, even when the device is asleep.

2.4 Developer resources

SDK: Our SDK wraps around complex Android libraries such as CameraX, abstracting some of the complexity away. It also includes a suite of sample apps that allow users to quickly set up on-device machine learning pipelines or live agents. Furthermore, we provided an example LiveKit application. LiveKit is a real-time communication platform for streaming audio, video, and data, which we use to connect the wearable to a live agent⁴. This enables users with no Android experience to set up a live agent on the wearable by simply cloning the application on the device and setting up the environment variables. This allows users to fine-tune their AI agent functionality with tool calls, system prompts, etc., without having to deal with processing input and output from the devices.

Documentation: Our developer documentation provides comprehensive content from setting up the device to accessing common libraries like camera, microphone etc. These resources, along with links to our SDK⁵, Discord server and GitHub community organisation⁶, can be accessed via our developer page⁷.

¹<https://source.android.com/license>

²<https://gs.statcounter.com/os-market-share/mobile/worldwide>

³<https://developer.android.com/media/camera/camerax>

⁴<https://docs.livekit.io/>

⁵<https://github.com/aisee-ai/starter-codebase.git>

⁶<https://github.com/aisee-community>

⁷<https://docs.aisee.ai>

3 User Study

3.1 Participants

We recruited 10 participants (6 male, 4 female; age: 23-45) who were computing professionals interested in developing heads-up computing applications. Participants came from various programming backgrounds and included Android developers, AI application engineers, and HCI Researchers (see supplementary Appendix A for a detailed participant breakdown).

3.2 Procedure & Data Collection

The study was run similar to a group workshop. We began with an introduction to the platform, covering the device hardware, operating system, and available I/O. We then delivered a technical demonstration to illustrate how to use the SDK and develop applications for the device. After the demo, participants completed a 6-hour development sprint (including a 1-hour lunch break) to build an application of their choice (Figure 1C). Following the sprint, we held a demo session where participants presented their applications and shared feedback.

Finally, participants were asked to complete System Usability Scale (SUS) [4] questionnaire to capture their perceived usability [15] and Developer Experience Index (DEXI) questionnaire to assess developer experience [12] (see supplementary appendix B). In addition, each participant took part in a semi-structured interview to elicit details about what they built, their development experiences, and suggestions for platform improvements. Interviews were recorded and subsequently analysed using thematic analysis [2].

4 Findings

4.1 System Usability & Developer Experience

The SUS scores showed a mean of 61.25 (SD = 15.78), with a 95% CI of [49.96, 72.54] which shows the platform was perceived as acceptable [5]. High scores for questions 5 (*I found the various functions in this system were well integrated*) and 6 (*I thought there was a lot of consistency in the system*) show that users believed that the system was consistent and well integrated.

The overall DEXI score had a mean of 5.21 (SD = 0.73), with a 95% CI of [4.69, 5.73] on the 1 to 7 scale, indicating a generally positive developer experience. (*The DEXI subscales and scoring procedure are summarised in the supplementary appendix B*). The Hedonic Quality (HQ) sub-scale, which covers aspects like motivating, supports creativity and enjoyable was 5.08/7, showing that our platform supported them cognitively and emotionally. Furthermore the General quality or recommendability score was 5.8/7 showing that despite certain usability discomfort users are willing to recommend the platform to others.

A notable pattern emerged by experience level where participants with Android development proficiency (P5–P7; see supplementary Appendix B) reported higher perceived usability and developer experience (SUS = 71.67, DEXI = 5.97) compared to the average values.

4.2 Range of projects

The projects developed during the sprint fell into three categories based on implementation approach: Android-only, Android with a

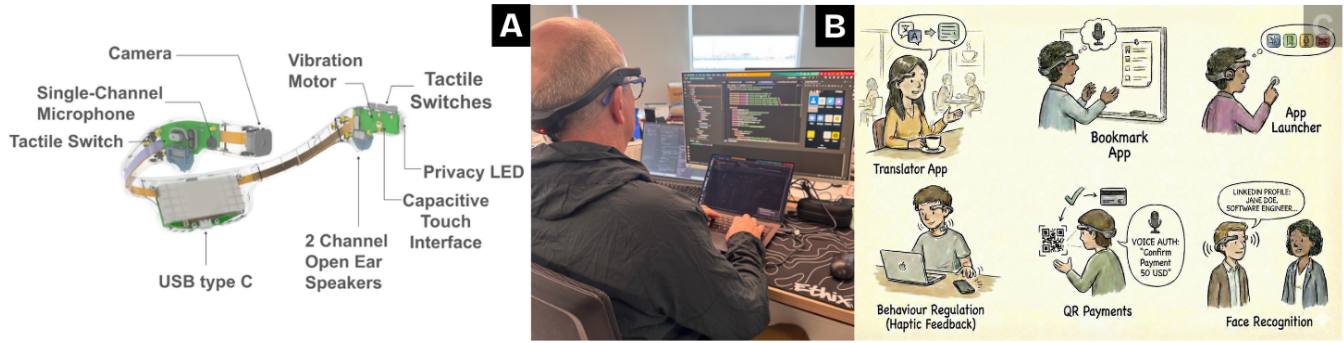


Figure 2: A. Hardware overview. B. Participant working with the headset. C. Applications created by participants.

custom backend, and Python-only using our provided starter application. We describe a selected subset of these projects (Figure 2C) below.

Android-only projects. The Android-only projects included: (1) a translator app that automatically detects the spoken language and performs translation; (2) a bookmarking app that uses the camera and user instructions to bookmark scenes, record an audio description, and review past bookmarks; (3) a gesture-detection app for controlling slides; and (4) an app launcher tailored to our platform for browsing and opening installed applications. The gesture-detection app ran real-time inference locally on the device. On a commercial device like Meta Ray-Ban, the same pipeline would require sending the video stream to a companion phone or to the cloud for processing, adding communication latency.

Android with a custom backend. Some applications developed during the workshop consisted of an Android application running on the headset, with an external backend integration for handling specific tasks. One example was a behaviour regulation app that used a locally-hosted Llava⁸ LLM to monitor a user’s phone usage and provide haptic feedback via the headset as a deterrent. Interestingly, the participant who developed the app noted that while “the headset was powerful enough to run the model on-device”, they opted to serve it from their computer due to time constraints of the workshop.

Python-only using the starter app. A project in this category utilised our provided LiveKit starter Android application, which handled all on-device Android operations, allowing development to occur exclusively in Python. The system functioned as a face recognition agent designed to detect a person and retrieve their LinkedIn information. This approach enabled the developer to bypass Android-specific coding to focus on the agent’s logic, as P9 explained: “I do not want to spend time coding on Android; this application would help me to work only on the agent and improve its functionality.”

Overall, the diversity of projects demonstrates the flexibility of our platform and highlights its potential for heads-up computing.

4.3 Other Insights

Hardware insights: Participants appreciated the form factor, with one participant mentioning, “I think this form factor is more useful for these applications than smart glasses, as it doesn’t have a screen that would distract users” (P10). This finding supports design decisions taken when developing our novel form-factor for screen free heads-up computing.

Several participants wanted the option to attach additional sensors, highlighting a need for extensibility. A concern raised was the battery life as some devices ran out of battery towards the end of the sprint. However, because the device runs Android, participants could continue testing on emulators while the device recharged.

Software stack and tooling insights: Participants proficient in Android were able to get started quickly and complete their applications for the device. Some of these participants mentioned that the lack of Google Play services on the device was a constraint for them.

Participants with less Android experience however spent considerable time setting up their Android environments. For example, the first participant to finish their project did it in 1.5 hours; however, 30 minutes of that was spent in setting up the development environment (P2). However, once the setup was completed, they too were able to build their applications quickly. A key reason was the provided sample applications and documentation, with one participant saying “It was much faster to build using the template apps rather than starting from scratch” (P1). This participant initially started building from scratch, and after 2 hours the system was not usable; however, after switching to the provided app and building on top of it they were able to get it working to a satisfactory state in 1 hour.

Agentic coding tools were used by most users. Some participants found them highly effective, with one participant saying, “It went perfectly; I got it working in one hour” (P2). while others reported that the tools were not of much use.

A strong takeaway was that developers were generally satisfied with the experience, with one participant noting, “I really enjoyed developing with the AiSee” (P7). After the workshop, several projects were submitted to our GitHub community⁹, suggesting sustained interest beyond the event.

⁸<https://ollama.com/library/llava>

⁹<https://github.com/aisee-community>

5 Conclusion

While our work demonstrates the potential of a screen-free, head-worn prototyping platform for heads-up computing, we identify several limitations in both the platform design and the study setup. Primary barriers for faster adoption included, onboarding friction for developers with limited Android experience, constraints from battery life during longer sprints, and platform limitations such as the lack of Google Play services and a modular sensor interface. Participants also requested more practical documentation examples (notably for camera streams) and expressed interest in expanding the device through additional sensors. Because our workshop captured outcomes over a short period, a longitudinal study is needed to understand longer-term benefits, challenges, and sustained usefulness in real-world settings. Future iterations of the platform will address the identified limitations through improvements in hardware modularity, software tooling, and developer resources. Overall, this work represents an initial step toward open, extensible, and developer-friendly infrastructures for screen-free heads-up computing, and provides insights that can inform the design of future wearable prototyping platforms in HCI.

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